

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – MINI APPLICATION DEVELOPMENT (CHATBOT /
SUMMARIZER / TRANSLATOR)

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 1-Mini Application Development (Chatbot / Summarizer / Translator)
Weightage:	20% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- **Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.**
- Implement the solution using **Python**.
- Select and justify an appropriate AI/LLM model or technique.
- Develop a **working application with a user interface**.
- Test the application using multiple input cases.
- Demonstrate handling of invalid, empty, or unexpected inputs.
- Clearly explain the application architecture and workflow.
- Submit the source code, test results, screenshots/demo, and documentation.
- Explain the limitations of the selected model/technique and suggest possible improvements.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

SECTION A – MINI APPLICATION DEVELOPMENT (CHATBOT / SUMMARIZER / TRANSLATOR)

AI CHATBOT APPLICATIONS

1. College Admission Chatbot – Develop an AI-powered chatbot that answers queries related to college admission, courses, eligibility, fees, and admission procedures. Provide a user-friendly interface and demonstrate the chatbot using at least 15 test queries.
2. Student Academic Support Chatbot – Develop a chatbot that assists students with questions related to subjects, assignments, examinations, attendance, and academic schedules. The chatbot should provide meaningful responses for different types of student queries.
3. Library Assistant Chatbot – Develop an AI chatbot for a college library that answers questions about book availability, borrowing rules, working hours, membership, and return policies. Include suitable fallback responses for questions outside the available knowledge.
4. Department Information Chatbot – Build a chatbot that provides information about a department, including faculty, programs, laboratories, events, facilities, and contact information. Demonstrate the chatbot using direct and paraphrased queries.
5. Campus Helpdesk Chatbot – Develop an AI-powered campus helpdesk chatbot that assists students with queries related to classrooms, laboratories, transport, facilities, events, and administrative services.
6. Course Information Chatbot – Develop a chatbot that answers questions about a selected course, including syllabus, units, learning outcomes, prerequisites, assessment methods, and laboratory activities.
7. Examination Support Chatbot – Build a chatbot that answers student queries about examination procedures, examination rules, registration, hall tickets, internal assessments, and result-related information.
8. Placement Assistance Chatbot – Develop a chatbot that provides information about placement eligibility, recruitment procedures, interview preparation, aptitude tests, resumes, and placement guidelines.
9. Technical Support Chatbot – Develop a chatbot that assists users with common technical problems such as password issues, software installation, login problems, network problems, and application errors. The chatbot should provide step-by-step solutions.
10. FAQ-Based Generative AI Chatbot – Create an AI chatbot using a collection of at least 30 frequently asked questions and answers. The chatbot should understand direct and paraphrased questions and generate appropriate responses.
11. Document-Based Chatbot – Develop a chatbot that answers questions from a provided document such as a college handbook, course regulations, or student guidelines. The system should respond based on the document and indicate when information is unavailable.
12. RAG-Based College Chatbot – Develop a Retrieval-Augmented Generation chatbot using a collection of college-related documents. The system should retrieve relevant information and generate answers based on the retrieved content.
13. Multilingual Student Chatbot – Develop a chatbot that accepts student questions in English and at least one Indian language and provides appropriate responses. Demonstrate the system using different types of queries.

14. Customer Support Chatbot – Develop an AI chatbot for a hypothetical e-commerce company that handles queries related to orders, delivery, returns, refunds, and product information. Include suitable fallback handling.

TEXT SUMMARIZATION APPLICATIONS

15. News Article Summarizer – Develop a text summarization application that accepts a long news article and generates a concise summary. Provide a simple interface and allow the user to control the approximate summary length.

16. Student Notes Summarizer – Build an application that summarizes lengthy study notes into important points. The system should accept text as input and generate a concise summary suitable for examination revision.

17. Research Paper Summarizer – Develop a summarization application for research papers. The system should generate a summary containing important information such as the problem, methodology, findings, and conclusion.

18. Meeting Minutes Summarizer – Develop an AI application that summarizes meeting transcripts and identifies important decisions, discussions, and action items.

19. Article Summarizer with Key Points – Create an application that accepts a long article and generates both a paragraph summary and a list of important key points.

20. Educational Content Summarizer – Develop a summarization application for educational materials. The system should convert lengthy learning content into a concise summary while preserving important concepts and terminology.

21. Email Summarizer – Build an AI application that accepts lengthy emails and generates a short summary containing the main purpose, important information, and required actions.

22. Multi-Document Summarizer – Develop an application that accepts multiple related documents and generates a combined summary highlighting common and important information from all documents.

23. Extractive vs Abstractive Summarizer – Develop an application that implements both extractive and abstractive summarization. Allow the user to enter a document and compare the summaries generated by the two approaches.

24. Summarizer with Length Control – Develop an AI summarization application that allows users to select summary length such as short, medium, or detailed. Demonstrate how the output changes for the same input text.

25. PDF Document Summarizer – Develop an application that accepts a PDF document, extracts its text, and generates a concise summary. Include appropriate handling for empty or unreadable documents.

26. Domain-Specific Summarizer – Select a domain such as education, finance, technology, or business and develop a summarization application specifically designed for that domain.

27. Summarization with Keyword Extraction – Develop an application that generates a summary along with the most important keywords from the input document.

MACHINE TRANSLATION APPLICATIONS

28. English-to-Tamil Translation Application – Develop a machine translation application that translates English text into Tamil. Provide a user-friendly interface and demonstrate the system using different types of sentences.

29. Tamil-to-English Translation Application – Develop an application that translates Tamil text into English. Test the application using simple, complex, and conversational sentences.

30. English-to-Indian-Language Translator – Develop a translation application that translates English text into one selected Indian language. Include suitable handling for unsupported or empty input.

31. Multilingual Translation Application – Develop an application that accepts text in English and translates it into at least two different target languages. Provide a language-selection option in the interface.

32. Translation with Language Detection – Develop a multilingual application that automatically detects the input language and translates the text into a user-selected target language.

33. Educational Content Translator – Build a translation application specifically for educational content. The system should translate short lessons, instructions, or study materials while preserving important technical terminology.
34. Chat Translation Application – Develop a simple multilingual chat application where users can enter messages in one language and receive the translated version in another language.
35. Document Translation Application – Develop an application that accepts a text document and translates its content into a selected target language. Provide an option to display or save the translated output.

INTEGRATED MINI APPLICATIONS

36. Chatbot + Summarizer Application – Develop an AI application that allows users to upload or enter a long document. The system should generate a summary and allow users to ask questions about the summarized content through a chatbot interface.
37. Translator + Chatbot Application – Develop a multilingual AI assistant that accepts user questions in one language, translates the input when required, generates an appropriate response, and displays the response in the user's selected language.
38. Summarizer + Translator Application – Develop an application that accepts a lengthy document, generates a concise summary, and translates the summary into a user-selected language.
39. Multilingual Document Assistant – Develop an integrated application that accepts a document, generates its summary, allows the user to ask questions about the document, and provides answers in a selected language.
40. Intelligent AI Assistant – Chatbot, Summarizer and Translator – Develop a complete AI application integrating chatbot, text summarization, and machine translation capabilities. The application should allow the user to select the required task, process the input appropriately, and display the result through a polished user interface. Demonstrate the system using multiple test cases and discuss the limitations and possible improvements.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Model/Technique Fit	30%	Correct, appropriate model/technique choice for the chosen application	Mostly appropriate choices made	Some inappropriate technique choices	Technique choice fundamentally flawed
Functional Completeness	25%	Application fully functional; solves a real use-case effectively	Mostly functional application	Partially functional application	Application is non-functional
Robustness & Reliability	20%	Handles edge cases; stable, reliable performance	Handles common cases well	Limited robustness to edge cases	Frequent failures during use
UI / Demo Polish	15%	Polished UI/demo with an engaging walkthrough	Clear demo, minor polish issues	Basic or rough demo	No usable demo presented

Design & Usage Documentation	10%	Comprehensive documentation of design & usage	Adequate documentation	Minimal documentation	No documentation provided
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Marks Evaluation:

Question No.	Model/Technique Fit (30%)	Functional Completeness (25%)	Robustness & Reliability (20%)	UI / Demo Polish (15%)	Design & Usage Documentation(10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						